GPS-only results from the report:

"RTK Positioning with Long Reference Data Outages"

2005-07-18, Trimble (U. Vollath, S. Klose)

The positioning performance was assessed for reference station delay (=outage) times between 1 and 500 seconds with the exception of the precise clock processing that can only be done for multiples of 300 seconds.

The different variants tested are in detail:

1. Take the outdated reference station data "as is". No corrections are applied in addition to what's done in the baseline processor assuming synchronous data. As the increase of the horizontal RMS (HRMS) is extremely linear, the main contributor was assumed to be unmodeled satellite clock errors, a process that is linear to a certain extend. This lead naturally to the next variant.
2. Apply the broadcast satellite clock error model to the reference and rover data. This is not done in the retail version of the processor as the satellite clock errors cancel out by single differencing between reference and rover for synchronous data. This is equivalent to apply the change in the satellite clock errors during the time span between receiving the reference data epoch and the rover data epoch. A very linear-looking term remained.
3. In addition to the satellite clock error, apply the general relativistic correction to reference and rover data caused by the eccentricity of the satellite orbits and precisely known using a simple formula. As with the previous method, the retail version dealing with synchronous data only doesn't require this correction. This had a surprisingly large effect on the accuracy level achieved. The increase with outage time is still very linear.
4. When using some kind of infrastructure software (e.g. GPSNet) ultra-rapid orbits and clocks are used to provide the best possible orbit information available at the time of the real-time positioning. The quality is much higher than that of the broadcast orbits. The aim was to eliminate the orbit (and satellite clock errors) as much as possible in real-time. The linear part of the error was largely reduced, and a horizontal accuracy of 10 cm with 3 minutes of reference station data outage was achieved.
5. To check for the maximum accuracy possible with proper modeling of satellite orbits, the previous experiment was repeated with final precise orbits, a product usually not available before one to two weeks after the mission. To explain the error budget, this information was valuable, though. The impact was not as big as anticipated.
6. After logically excluding the common suspects as tropospheric errors (ionosphere was eliminated by using the ionosphere free carrier phase combination), an error increase over time remained with a linear characteristics at several 100 seconds of

delay. As the error plots showed a strange time dependence and periodicity, the final processing mode was implemented. Here, the final precise clocks are used that are available at a data rate of 300 seconds in contrast to the precise orbits provided every 900 seconds only. The higher data rate allows targeting an analysis at the most interesting time delay of 300 seconds where precise reference and rover clocks are available at the matching times.

Figure 1 presents the results for the variants described. For numerical results please see Section 7.

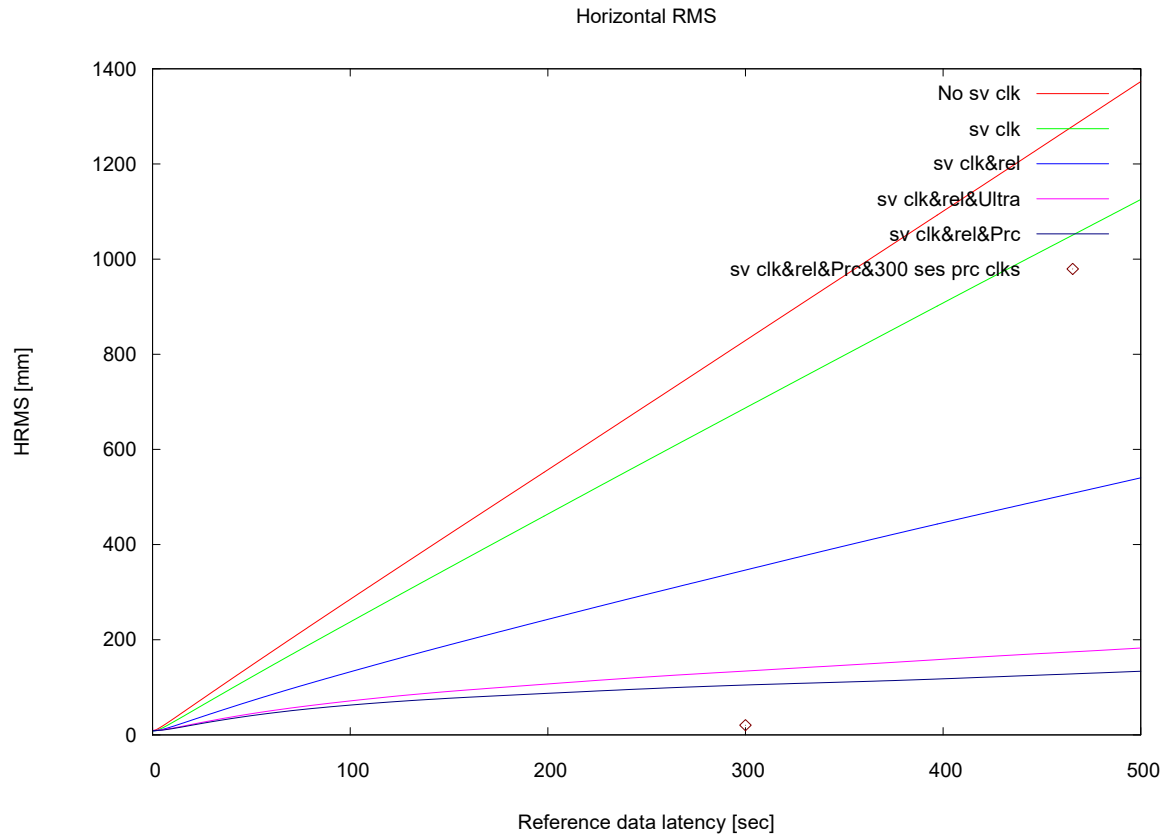


Figure 1: Horizontal RMS degradation for different reference outage time spans and corrections

The most remarkable entry is the small point at 300 seconds outage time for the precise 300 seconds clocks at about 20 mm horizontal RMS. This is still in the accuracy level of RTK kinematic positioning using a functioning reference data link.

The results show that using “text book” modeling the accuracies for positioning can be retained for a surprisingly long time. However, the final limiting factor has been shown indirectly (by removing it) to be the satellite clock errors.

Another test supports this conjecture. Figure 2 shows the difference between a linearly interpolated clock error using two epochs 300 seconds before respective after the epoch shown and the value for that epoch retrieved from the IGS precise clock product. The error is actually in the range needed to explain the observed position errors. The atomic clocks on the satellites have excellent long term stability yet there is an additional component that dominates the reference station data latency problem.

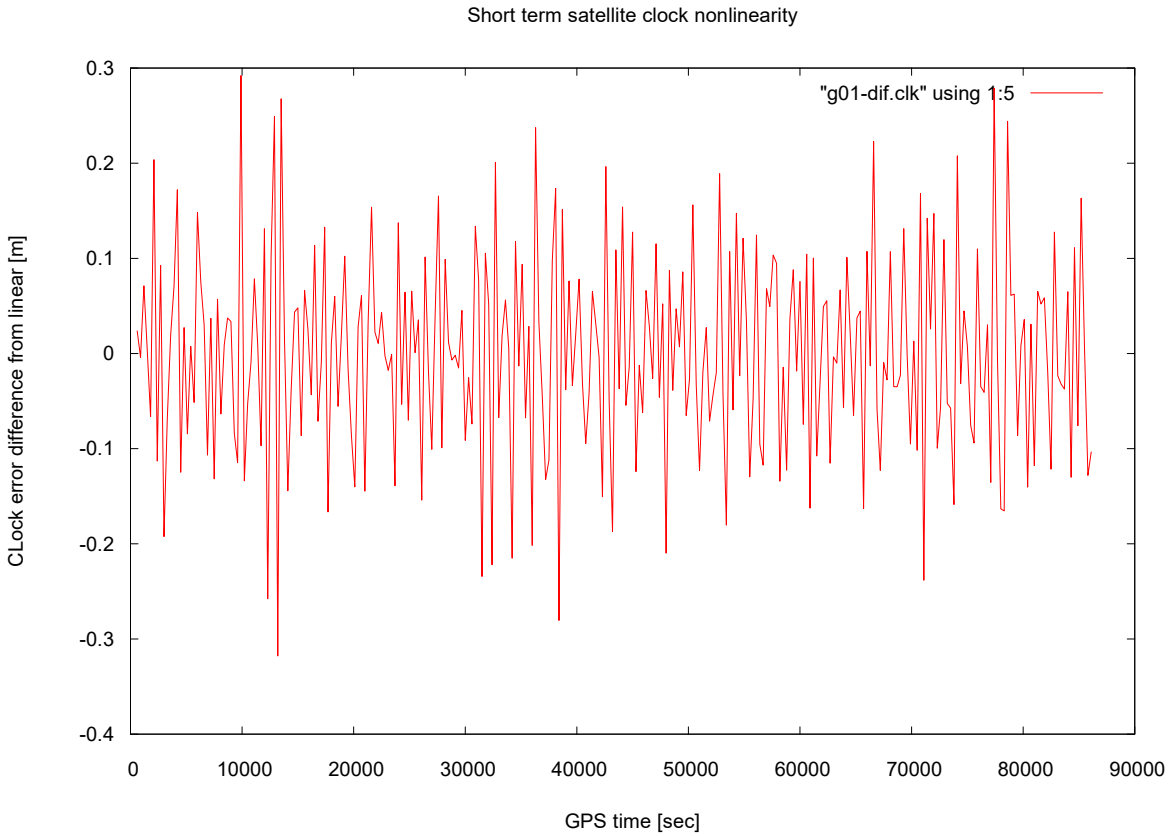


Figure 2: Difference between 600 seconds interpolation and precise IGS clock value for GPS 01

This presumption had first surfaced at Terrasat a few years ago when linear modeling of satellite clock errors in the geometry filter failed due to additional unexplained few decimeter error components. If these findings hold as found in the data set used, it would be sufficient to provide a correction data stream that updates the clock error say every 10-20 seconds with a data range of a meter and a centimeter-level resolution. For example, 8 bits of data for every say 12 satellite in view with an assumed overhead of another 8 bits would be less than 200 bits of information to be updated.

This requires some additional information transmitted during the availability of the full reference data stream. Many other constellations of corrections based on broadcast and ultra-rapid orbits and information from VRS processing are possible.